A Local Einstein–Scalar BV/BRST Benchmark for Boundary Reconstruction Programs

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Abstract

Boundary and reconstruction approaches to quantum gravity often formulate strong requirements—positivity, gauge reduction, causal propagation, sewing, state selection, and semiclassical recovery—without exhibiting a single continuum sector in which the requirements can be checked with compatible domains and conventions. This paper constructs a deliberately narrower benchmark. We start from the Einstein–scalar action with negative cosmological constant, expand about the zero-scalar AdS_4 background, and work on a causally convex globally hyperbolic interior double cone whose closure is disjoint from the timelike conformal boundary. The classical nonminimal BV datum, its gauge-fixed tangent complex, and the exact de Donder ghost operator are fixed explicitly. Using source-specialized results on linearized Einstein–Klein–Gordon theory and a derived block inversion, we obtain a full advanced/retarded Green matrix for $(h, \phi, c, \bar{c}, b)$ for every real auxiliary gauge parameter, including the nonschur value zero.

For a refined double cone we embed the local geometry into a spatially compact Fuchsian locally-AdS host. Published Hadamard existence theorems, a Fredholm left-inverse construction, and an explicit normalization transport then yield a five-field BRST-compatible Hadamard system and a positive, no-extra-null ghost-number-zero one-particle space. The free functional algebra is built on the equicausal, rather than unrestricted microcausal, domain. It satisfies the exact graded Ward identity, Einstein causality, and a fixed-ambient time-slice property. We also prove a sharp limitation: the ordinary field-algebra dagger does not descend to the same BRST cohomology in the frozen nonminimal reality convention. Finally, the first $h\phi\phi$ matter deformation and its scalar-antifield completion form an exact classical first-order BV subcocycle; a conserved renormalized one-insertion stress normalization is available, while the associated local AMWI/QME obstruction class is typed but not computed.

The construction is a benchmark, not a derivation of gravity. It assumes the Einstein–scalar action. It supplies neither global reflecting AdS boundary conditions, a full interacting quantum master equation, an operational boundary instrument, a regulator-removal theorem, nor an emergent infrared bridge. The paper separates source-specialized inputs from project-level compositions throughout and states an exact stop line for further promotion.

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1 Introduction

1.1 Why a benchmark is needed

The phrase "boundary reconstruction" covers programs with rather different ontologies. Some begin with amplitudes assigned to codimension-one boundaries, some with nets of observables, some with constraint complexes, and some with positive functionals or moment data. Despite these differences, a mature continuum sector has to make several structures coexist: a gauge-reduced classical phase space, causal propagation, a quantum algebra, an admissible state, a physical positivity statement, and sufficiently controlled interactions. It is easy to verify each item in a different model or convention. It is much harder to exhibit them on one domain and to state precisely which step fails when one attempts to pass from a local effective field theory to a gravitational boundary theory.

The purpose of this paper is to provide such a controlled test object. We call it the *local gauge benchmark*. It is local, perturbative, and effective. Its starting point is not an unknown microscopic boundary amplitude but the familiar Einstein–scalar action. Consequently, success of the benchmark demonstrates compatibility of a reconstruction architecture with a standard local gravitational sector; it cannot be used to claim that Einstein gravity has been derived from that architecture.

The main technical issue is not writing the classical BV action. The difficulty is to keep the following chain typed and normalized:

$$\begin{aligned}\text{BV datum} \longrightarrow \text{Green matrix} \longrightarrow \text{Hadamard pair} \longrightarrow \text{equicausal star algebra} \\ \longrightarrow \text{BRST reduction} \longrightarrow \text{time slice} \longrightarrow \text{first interaction test.}\end{aligned} \tag{1}$$

Gauge fields, ghosts, antighosts, the Nakanishi–Lautrup field, source pairings, left/right derivatives, and the rescaling between geometric and source conventions all matter. A missing sign or an unjustified restriction of a covariance can break the Ward identity or the physical inner product.

1.2 Relation to established work

The classical functional BV framework and its locally covariant gravity specialization are due to Fredenhagen, Rejzner, Brunetti and collaborators [9, 14]. Gauge-invariant quantization of linearized Einstein–Klein–Gordon systems and linearized gravity on globally hyperbolic backgrounds was developed in [13, 20, 21]. Green-hyperbolic operators and their restriction properties are treated systematically in [2]. The BRST phase-space and Hadamard criteria used below come from [28]; the nontrivial gravity-state input is Gérard's existence theorem on Einstein spacetimes with compact Cauchy surfaces [16].

Two corrections to older pAQFT shorthand are essential. First, the traditional unrestricted class of microcausal functionals is not closed under the Peierls bracket. The equicausal domain introduced by Hawkins, Rejzner, and Visser repairs this defect and supports the star product and time-slice homotopy [22]. Second, a positive state on the reduced bosonic algebra is not by itself a full nonminimal BRST state. We therefore construct the tensor/vector Ward pair on a host, transport it through the full five-field Green matrix, and prove the local no-extra-null statement separately.

1.3 Results and provenance vocabulary

We use four provenance descriptions.

standard	A definition or immediate formal consequence, not an originality claim.
source-specialized	A published theorem is applied after its hypotheses and convention map are printed.
derived	A project-level block inversion, compatibility proof, or obstruction argument is supplied here.
open	A stated target whose required proof or construction is absent.

The principal derived contributions are the explicit full nonminimal Green matrix, the compact-host Fredholm/Ward completion adapted to the selected reduced state, the normalization map into the five-field local benchmark presentation, the local zero-extension injectivity argument, the dagger-descent counterexample, and the first matter BV subcocycle. The general BV, Green-hyperbolic, Hadamard, equicausal, and Epstein–Glaser mechanisms remain source results.

Scope ceiling. Every theorem below concerns a formal perturbative expansion on an interior globally hyperbolic region. There is no claim of a global AdS Hilbert space, a nonperturbative or ultraviolet-complete theory, an independently functorial physical net, or an observable boundary readout. The strongest interacting statement is a first classical matter subcocycle plus a typed but uncomputed quantum anomaly class.

2 Local geometric and dynamical datum

2.1 The interior domain

Let $(\widetilde{\mathrm{AdS}}_4, \bar{g})$ be the universal cover of four-dimensional anti-de Sitter spacetime with radius L and mostly-plus signature. In global coordinates,

$$\mathrm{d}s^2_{\bar{g}} = \frac{L^2}{\cos^2\rho}\left(-\mathrm{d}t^2 + \mathrm{d}\rho^2 + \sin^2\rho\,\mathrm{d}\Omega_2^2\right), \qquad \bar{R}_{\mu\nu} = -\frac{3}{L^2}\bar{g}_{\mu\nu}, \tag{2}$$

so that $\bar{R} = -12/L^2$, $\bar{C}_{\mu\nu\rho\sigma} = 0$, and $\Lambda = -3/L^2$.

Global AdS has a timelike conformal boundary and is not globally hyperbolic without a boundary prescription. We therefore choose a connected, causally convex, globally hyperbolic double cone U_F with compact closure contained in the interior of $\widetilde{\mathrm{AdS}}_4$. The closure is disjoint from the conformal boundary. All proper gauge parameters, test sections, switching functions, and functional derivatives used in integrations by parts have compact support in U_F.

This restriction is structural, not cosmetic. It makes the standard advanced and retarded problem meaningful and removes conformal-boundary terms, but it also means that no conclusion below is a theorem about reflecting global AdS.

2.2 Einstein–scalar action and background

Fix

$$\kappa^2 = 32\pi G, \qquad m^2L^2 = -2, \qquad \bar{\phi} = 0, \tag{3}$$

and use the standard-quantization label $\Delta_+ = 2$ for later comparison with an AdS boundary problem. The mass lies above the AdS_4 Breitenlohner–Freedman bound $m^2L^2 \geq -9/4$ [7]. On the local interior domain the label Δ_+ does not impose a boundary condition.

The classical action is

$$\begin{aligned} S_{\mathrm{cl}}[g,\phi] = {} & \frac{2}{\kappa^2}\int_{U_F}(R(g) - 2\Lambda)\,\mathrm{d}\mu_g \\ & - \frac{1}{2}\int_{U_F}\left(g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi + m^2\phi^2\right)\mathrm{d}\mu_g. \end{aligned} \tag{4}$$

Compactly supported variation gives

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{\kappa^2}{4}T_{\mu\nu}, \qquad (\Box_g - m^2)\phi = 0, \tag{5}$$

where

$$T_{\mu\nu} = \nabla_\mu\phi\nabla_\nu\phi - \frac{1}{2}g_{\mu\nu}((\nabla\phi)^2 + m^2\phi^2). \tag{6}$$

The background $(\bar{g}, 0)$ solves equation (5).

Write

$$g = \bar{g} + \kappa h. \tag{7}$$

At free order the scalar block is

$$P_\phi = \bar{\Box} - m^2 = \bar{\Box} + \frac{2}{L^2}. \tag{8}$$

The normalization in equation (7) is retained throughout: a canonical graviton backreaction of order $\kappa\hbar$ corresponds to a physical metric shift of order $\kappa^2\hbar$.

3 Classical BV complex and free tangent theory

3.1 Nonminimal fields and antibracket

The nonminimal field multiplet is

$$\Phi = (g_{\mu\nu}, \phi, c^\mu, \bar{c}_\mu, b_\mu), \tag{9}$$

with ghost numbers $0, 0, 1, -1, 0$ and parities even, even, odd, odd, even. For every field Φ^A introduce a density-valued antifield $\Phi^\ddagger_A$ of ghost number $-1 - \mathrm{gh}(\Phi^A)$ and opposite parity. Our antibracket is

$$(F, G) = \sum_A \int_{U_F} \left(\frac{\delta_r F}{\delta \Phi^A} \frac{\delta_l G}{\delta \Phi^\ddagger_A} - \frac{\delta_r F}{\delta \Phi^\ddagger_A} \frac{\delta_l G}{\delta \Phi^A} \right). \tag{10}$$

The BRST differential acts from the right Hamiltonian convention, $sF = (F, S_{\mathrm{BV}})$.

Freeze

$$sg = \mathcal{L}_c g, \quad s\phi = \mathcal{L}_c \phi, \quad sc = \frac{1}{2}[c, c]_{\mathrm{gr}}, \quad s\bar{c} = b, \quad sb = 0, \tag{11}$$

where $[c, c]^\mu_{\mathrm{gr}} = 2c^\nu \nabla_\nu c^\mu$. The nonminimal master action is

$$S_{\mathrm{BV}} = S_{\mathrm{cl}} + \int_{U_F} (g^{\ddagger\mu\nu} \mathcal{L}_c g_{\mu\nu} + \phi^\ddagger \mathcal{L}_c \phi + \tfrac{1}{2} c^\ddagger_\mu [c, c]^\mu_{\mathrm{gr}} + \bar{c}^{\ddagger\mu} b_\mu). \tag{12}$$

Theorem 3.1 (Classical master equation; source-specialized)**.** *Under the compact-support convention on U_F,*

$$(S_{\mathrm{BV}}, S_{\mathrm{BV}}) = 0, \qquad s^2 = 0. \tag{13}$$

Proof. Diffeomorphism invariance of equation (4) holds without a boundary term for compactly supported gauge parameters. Lie derivatives close under the graded vector-field bracket, whose graded Jacobi identity proves nilpotence on g, ϕ, c. The pair $(\bar{c}, b)$ is contractible. In the canonical antibracket these statements are equivalent to the classical master equation. This is the standard BV implication [6, 14] specialized to the present action. □

3.2 Gauge fixing and tangent complex

Rescale the full ghost by $c_{\mathrm{full}} = \kappa c$ and the corresponding antifields canonically. The free differential is

$$s_0 h_{\mu\nu} = 2\bar{\nabla}_{(\mu} c_{\nu)}, \qquad s_0 \phi = 0, \qquad s_0 c = 0, \qquad s_0 \bar{c} = b, \qquad s_0 b = 0. \tag{14}$$

Let

$$F_\mu(h) = \bar{\nabla}^\nu h_{\mu\nu} - \frac{1}{2} \bar{\nabla}_\mu h, \qquad \Psi = \int_{U_F} \bar{c}^\mu \left(F_\mu(h) + \frac{\lambda}{2} b_\mu \right) \mathrm{d}\mu_{\bar{g}}, \tag{15}$$

with real constant λ. If P_0 is the ungauge-fixed Euler–Lagrange Hessian and

$$K\xi = (2\bar{\nabla}_{(\mu} \xi_{\nu)}, 0), \tag{16}$$

then the compactly supported tangent complex is

$$\Gamma_c(TU_F) \xrightarrow{K} \Gamma_c(S^2 T^* U_F \oplus \mathbb{R}) \xrightarrow{P_0} \Gamma_c(S^2 T^* U_F \oplus \mathbb{R})^* \xrightarrow{K^*} \Gamma_c(TU_F)^*. \tag{17}$$

Gauge invariance gives $P_0 K = 0$ and $K^* P_0 = 0$.

With $[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma$, the de Donder ghost block is

$$(P_{\rm gh}c)_\mu = \bar{\Box}c_\mu + \bar{R}_\mu{}^\nu c_\nu = \left(\bar{\Box} - \frac{3}{L^2}\right)c_\mu. \tag{18}$$

At the principal-symbol level, $F(k)K(k)c = k^2 c$. This identity checks gauge fixing of pure-gauge directions; it is not a derivation of the complete lower-order graviton Hessian.

3.3 Free cocycles and their limited meaning

Because the background Weyl tensor and scalar vanish,

$$s_0 C^{(1)}[h] = 0, \qquad s_0\phi = 0. \tag{19}$$

Indeed, the gauge change of a linearized natural tensor is the Lie derivative of its background value. Compactly smeared linearized Weyl and scalar observables therefore give free ghost-number-zero cocycles. They do not classify all cohomology classes. Nor do they make the point fields $h_{\mu\nu}(x)$ or $\phi(x)$ nonlinear gravitational observables. A nonlinear relational dressing is a later, absent construction.

4 The full nonminimal Green matrix

4.1 Source variables and convention map

The normally hyperbolic source formulation of the linearized Einstein–Klein–Gordon system in [20] is particularly convenient. Let $\Theta = (\theta, \zeta)$ denote the trace-reversed source variables and define

$$\mathcal{A}(h,\phi) = \left(\frac{\kappa}{4}\left(h_{\mu\nu} - \frac{1}{2}\bar{g}_{\mu\nu}h\right), \frac{\kappa}{2}\phi\right). \tag{20}$$

Write $\bar{K}$ for the source gauge generator and $D = \bar{K}^\dagger$. On the zero-scalar Einstein background set

$$R = D\bar{K} = -\frac{1}{2}(\bar{\Box} + \mathrm{Ric}), \qquad Q = 2R, \qquad \widetilde{P} = \bar{P} + 2\bar{K}D. \tag{21}$$

Both $\widetilde{P}$ and R are normally hyperbolic up to the nonzero scalar factor in R. Let $E_\pm$ and $C_\pm$ be their retarded/advanced Green operators, with the sign convention $\operatorname{supp} G_\pm f \subset J^\pm(\operatorname{supp} f)$.

The source and local benchmark quadratic actions obey

$$P_{0,\mathrm{b}} = -\frac{4}{\kappa^2}\mathcal{A}^\dagger \bar{P}\mathcal{A}. \tag{22}$$

The minus sign is forced by the source action convention and reconciles the scalar equation $\bar{\Box} - m^2$ with the source block $-\bar{\Box} + m^2$.

4.2 Euler matrix

In field and Euler-row order $(H, c, \bar{c}, b)$, with $H = (h, \phi)$, the gauge-fixed free operator is

$$P^{\mathrm{b}}_{\mathrm{nm},\lambda} = \begin{pmatrix} P_0 & 0 & 0 & F^\dagger \\ 0 & 0 & M^\dagger & 0 \\ 0 & -M & 0 & 0 \\ F & 0 & 0 & \lambda I \end{pmatrix}, \qquad M = FK = \bar{\Box} + \mathrm{Ric}. \tag{23}$$

In grouped source variables $(\Theta, \beta; \eta, \bar{\eta})$ the even and odd blocks are

$$B_\lambda = \begin{pmatrix} -\bar{P} & \bar{K} \\ D & \lambda I \end{pmatrix}, \qquad O = \begin{pmatrix} 0 & -R^\dagger \\ R & 0 \end{pmatrix}. \tag{24}$$

The full matrix is graded formally self-adjoint. Antifields generate the BV differential but are not additional propagating rows of this field Hessian.

To define a supported square, let $C_{+,c}$ be the retarded map on compact sources and $C_{+,pc}$ its canonical extension to past-compact sections; similarly use $C_{-,c}$ and $C_{-,fc}$. Put

$$C_+^2 = C_{+,pc} C_{+,c}, \qquad C_-^2 = C_{-,fc} C_{-,c}. \tag{25}$$

Then $RC_\pm^2 = C_\pm$ and $C_\pm^2 R = C_\pm$ on compact sections, and the causal support orientation is preserved [2].

Theorem 4.1 (Full Green matrix for every real λ; derived)**.** *The even Green operators are*

$$G_\pm^{(B_\lambda)} = \begin{pmatrix} -E_\pm + \dfrac{1-2\lambda}{2} \bar{K} C_\pm^2 D & \bar{K} C_\pm \\ C_\pm D & 0 \end{pmatrix}, \tag{26}$$

and the odd Green operators are

$$G_\pm^{(O)} = \begin{pmatrix} 0 & C_\pm \\ -C_\pm & 0 \end{pmatrix}. \tag{27}$$

Together they are two-sided advanced/retarded Green operators for the source matrix. No division by λ occurs, so the statement includes $\lambda = 0$.

Proof. The gauge identities imply

$$\widetilde{P}\bar{K} = 2\bar{K}R, \qquad D\widetilde{P} = 2RD. \tag{28}$$

Uniqueness of supported Green solutions therefore gives

$$E_\pm \bar{K} = \frac{1}{2}\bar{K} C_\pm, \qquad DE_\pm = \frac{1}{2} C_\pm D, \qquad \bar{P}E_\pm = E_\pm \bar{P} = 1 - \bar{K} C_\pm D. \tag{29}$$

To solve $B_\lambda(\Theta, \beta) = (f, g)$, apply D to the first row. One obtains $R\beta = Df$, hence $\beta = C_\pm Df$. The second row gives $D\Theta = g - \lambda\beta$, and substitution yields

$$\widetilde{P}\Theta = -f + 2\bar{K}g + (1-2\lambda)\bar{K}\beta. \tag{30}$$

Applying $E_\pm$ and equation (29) produces equation (26). Direct left and right multiplication uses $RC_\pm^2 = C_\pm = C_\pm^2 R$; the residual off-diagonal coefficient is $-\frac{1}{2} + \frac{1-2\lambda}{2} + \lambda = 0$. Locality of $D, \bar{K}$ and transitivity of causal futures/pasts prove the support estimate. Equation equation (27) is the direct inverse of the off-diagonal normally hyperbolic block. □

Let

$$J_{\mathrm{nm}} = \mathrm{diag}\left(\mathcal{A}, \kappa, \frac{\kappa}{2}, -\frac{\kappa}{2}\right). \tag{31}$$

The local benchmark Green maps are

$$G_{\pm}^{(P^{\mathrm{b}}_{\mathrm{nm},\lambda})} = \frac{\kappa^2}{4} J_{\mathrm{nm}}^{-1} G_{\pm}^{(P^{H}_{\mathrm{nm},\lambda})} (J_{\mathrm{nm}}^{\dagger})^{-1}. \tag{32}$$

At $\lambda = 1/2$ the $C_{\pm}^2$ term vanishes. For $\lambda \neq 0$ one may eliminate b and obtain the equation operator

$$\bar{P}_\lambda = \bar{P} + \lambda^{-1} \bar{K} D, \qquad G_{\pm}^{(\bar{P}_\lambda)} = E_{\pm} + \frac{2\lambda - 1}{2} \bar{K} C_{\pm}^2 D. \tag{33}$$

At $\lambda = 0$ the Schur complement does not exist, although the full block Green operator remains well defined.

Remark 4.2 (Source boundary). Normal hyperbolicity of $\widetilde{P}$ and R, together with their source convention, is specialized from [20]. The four-by-four nonminimal matrix, equations (26) and (27), its $\lambda = 0$ treatment, and the exact local benchmark pullback are the derived composition. The result proves propagation only. It does not produce a state or a physical representation.

5 Reduced phase space and the Fuchsian Hadamard host

5.1 Gauge-invariant test quotient

Use the source pairing

$$\langle (\theta, \zeta), (\theta', \zeta') \rangle_F = \int_{U_F} \left(4\theta^{\mu\nu} \theta'_{\mu\nu} - 2 \operatorname{tr} \theta \operatorname{tr} \theta' + \zeta \zeta' \right) \mathrm{d}\mu_{\bar{g}}. \tag{34}$$

The compact-test quotient is

$$\mathcal{E}^H_{U_F} = \frac{\operatorname{Ker}(D : \Gamma_c(\mathcal{V}) \to \Gamma_c(TU_F))}{\bar{P}\Gamma_c(\mathcal{V})}, \tag{35}$$

where $\mathcal{V} = S^2 T^* U_F \oplus \mathbb{R}$. At $\bar{\phi} = 0$ it splits algebraically into graviton and scalar summands. If $G = G_+ - G_-$ is the reduced causal propagator,

$$\sigma_H([f], [g]) = \langle f, Gg \rangle_F \tag{36}$$

is a well-defined presymplectic form. The local benchmark form is

$$\sigma_{\mathrm{b}}(f, g) = \frac{\kappa^2}{4} \sigma_H(J^\vee f, J^\vee g), \qquad J^\vee = (J^*)^{-1}. \tag{37}$$

Equivalently, the identified commutators use $\hbar_H = (\kappa^2/4)\hbar_{\mathrm{b}}$.

The quotient defines both a polynomial CCR algebra and, separately, the Weyl C^*-algebra

$$W(f)W(g) = e^{-\frac{i}{2}\hbar_{\mathrm{b}}\sigma_{\mathrm{b}}(f,g)} W(f+g), \qquad W(f)^* = W(-f). \tag{38}$$

Degeneracy of σ produces central Weyl elements but does not obstruct the universal construction. Causal support yields Einstein commutativity for disjoint compact representatives. Standard Green-hyperbolic deformation gives a fixed-ambient time-slice statement. These are source-specialized consequences of [13, 20, 21], not new general CCR theorems.

5.2 A compact-Cauchy locally-AdS host

The positive state is constructed on a refined domain. A Fuchsian AdS region has the form

$$E_v \simeq (-\pi/2, \pi/2) \times \mathbb{H}^3, \qquad g_E = -L^2 \mathrm{d}\tau^2 + \cos^2 \tau \, h_{\mathbb{H}^3, L}. \tag{39}$$

Choose U_F with compact closure $K \Subset E_v$. Let Γ_0 be a torsion-free cocompact lattice in $SO_0(1,3)$. Only finitely many nonidentity translates γK can be causally related to K. Residual finiteness permits passage to a finite-index normal subgroup Γ excluding this finite bad set. The quotient

$$M_\Gamma = \Gamma \backslash E_v \tag{40}$$

is a four-dimensional globally hyperbolic Einstein spacetime with compact Cauchy surface $\Gamma\backslash\mathbb{H}^3$, and the quotient map restricts to a causally convex isometric embedding $i : U_F \hookrightarrow M_\Gamma$. The geometric input belongs to the Fuchsian AdS theory surveyed in [3]; the finite bad-set refinement is the derived host-selection argument.

5.3 Reduced positive Hadamard state

Gérard's theorem applies to M_Γ because it is four-dimensional, globally hyperbolic, Einstein, and has compact Cauchy surfaces [16]. Let $\Lambda_G^\pm$ be the resulting positive gauge-invariant graviton Hadamard covariances. The source convention uses a constant trace-reversal bridge $T_g = 2I$. Its physical pairing is a factor-two homothety, so define

$$\Lambda_{H,g}^\pm(q, r) = \frac{1}{2} \Lambda_G^\pm(T_g q, T_g r). \tag{41}$$

Choose scalar Klein–Gordon Hadamard covariances on the same host, whose existence is standard [17, 26]. Since the free blocks decouple at $\bar{\phi} = 0$, take the direct sum and pull it back by zero extension:

$$\Lambda_{H,U_F}^\pm(q, r) = \Lambda_{H,M_\Gamma}^\pm(i_* q, i_* r). \tag{42}$$

Finally set

$$\Lambda_{\mathrm{b},U_F}^\pm(f, g) = \hbar_H \Lambda_{H,U_F}^\pm(J^\vee f, J^\vee g). \tag{43}$$

Assumption 5.1 (Imported physical Hadamard state)**.** Use the positive gauge-invariant graviton state on the compact-Cauchy Einstein host and a scalar Hadamard state, with the normalization

$$\Lambda_{\mathrm{b},U_F}^+ - \Lambda_{\mathrm{b},U_F}^- = i\hbar_{\mathrm{b}} \sigma_{\mathrm{b}}. \tag{44}$$

Existence in the present decoupled Einstein-host setting follows from the cited source results and the causal restriction map. A separate companion construction proves the coupled canonical Einstein–scalar physical Hadamard theorem for a finite fixed positive scalar target on compact Cauchy surfaces, under a vanishing potential-gradient condition along the on-shell background image [1]. That theorem is cited rather than re-proved here. The formulas above record the normalization required to use its compatible reduced state in this benchmark; no claim of a second new state-existence theorem is made. In particular, a physical-state theorem alone does not provide the nonminimal Ward pair, same-complex dagger or interacting anomaly calculation.

Restriction need not preserve purity or faithfulness on central Weyl elements. A full nonminimal BRST covariance requires more than this theorem; that construction is the subject of the next section.

6 Conditional Ward completion and nonminimal transport

The following composition uses the declared compact-host Fredholm and Hadamard inputs. Its new content is the explicit left-inverse/Ward-repair and convention transport algebra. The general microlocal and compact-test quotient machinery is imported from the cited results, including the companion physical-state theorem; finite coefficient checks are not proofs of those analytic inputs.

6.1 Cauchy complex and a tame left inverse

Let Σ_Γ be a compact Cauchy surface of M_Γ. In Gérard's adapted Sobolev scale write

$$\mathsf{H}_j^s = H^s(\Sigma_\Gamma; V_j) \oplus H^{s-1}(\Sigma_\Gamma; V_j), \tag{45}$$

and denote the Cauchy gauge map by

$$k = K_\Sigma : \mathsf{H}_1^{s+1} \longrightarrow \mathsf{H}_2^s. \tag{46}$$

Gérard's gauge-fixing map gives an operator R_Σ such that $A = R_\Sigma k$ is Fredholm of index zero and has an elliptic first-order diagonal part plus a smoothing remainder [16]. Choose a pseudodifferential generalized inverse G with

$$GA = 1 - \pi_F, \qquad F = \operatorname{Ker} A. \tag{47}$$

For the chosen Fuchsian quotient, $\operatorname{Ker} k = 0$. A global Killing one-form lifts to a Γ-invariant element of $\mathfrak{so}(2,3)$. Borel density reduces invariance to the centralizer of the embedded $\mathfrak{so}(1,3)$, and that centralizer is zero. Let $E = kF$, let $k_F : F \to E$ be the resulting isomorphism, and choose a smoothing projection π_E onto E. Define

$$L = GR_\Sigma + k_F^{-1}\pi_E(1 - kGR_\Sigma). \tag{48}$$

The inverse k_F^{-1} is applied only after π_E.

Proposition 6.1 (Fredholm left inverse; derived)**.** *The operator equation (48) obeys*

$$Lk = 1, \qquad Q_B = kL, \qquad Q_B^2 = Q_B, \qquad \operatorname{Ran} Q_B = \operatorname{Ran} k =: B. \tag{49}$$

It is of order -1 in the adapted scale up to a finite-rank smoothing term. Hence B is closed and complemented at every Sobolev order, with a compatible smooth Fréchet splitting.

Proof. For $x \in \mathsf{H}_1^{s+1}$, $GR_\Sigma kx = (1 - \pi_F)x$, and $(1 - kGR_\Sigma)kx = k\pi_F x \in E$. Substitution into equation (48) gives $Lkx = x$. The remaining identities follow immediately. Pseudodifferential order and finite-rank smoothness follow from the selected parametrix and projections. The estimates are inherited uniformly through the compatible Sobolev scale. □

Let q_j denote the nondegenerate Cauchy charges, interpreted as dualities $\mathsf{H}_j^s \times \mathsf{H}_j^{1-s} \to \mathbb{C}$. Put

$$C = \operatorname{Ker} k^\dagger, \qquad B = \operatorname{Ran} k \subset C. \tag{50}$$

The charge adjoint of $Lk = 1$ gives $k^\dagger L^\dagger = 1$. Consequently

$$Q_C = 1 - L^\dagger k^\dagger \tag{51}$$

projects onto C, and its dual-scale adjoint has kernel B. Therefore

$$C^{\perp q_2} = B. \tag{52}$$

The use of dual Sobolev orders is essential; no same-order adjoint is silently assumed.

6.2 Retaining the selected physical state

Gérard's regular and singular constructions provide a quotient selector p on C such that

$$pc = c \mod B, \qquad pk = 0, \qquad C = B \oplus P, \qquad P = \mathrm{Ran}(p|_C). \tag{53}$$

In the singular branch p contains a finite-rank correction and therefore retains, rather than discards, the finite-dimensional singular physical sector.

Let $\mu_2^{\pm}$ and $\mu_1^{\pm}$ be the unmodified tensor and vector Hadamard pseudocovariances. The desired reduced tensor state and the restriction of μ_2 to C/B have the same causal difference. Their common difference d_C is smoothing. Extend it to all tensor Cauchy data by

$$a_d(x, y) = d_C(Q_C x, Q_C y). \tag{54}$$

Adding the evolved bisolution associated with a_d equally to the two tensor signs preserves the causal difference and Hadamard wavefront set while making the restriction to $C \times C$ exactly the selected reduced state.

The tensor/vector Ward relation may initially hold only modulo a common smoothing defect

$$S = k\widehat{\lambda}_1^{\pm} - \widehat{\lambda}_2'^{\pm} k. \tag{55}$$

For every $c \in C$ and x in the vector Cauchy space, $q_2(c, Sx) = 0$. By equation (52), $Sx \in B$. Define

$$F = SL, \qquad A_2 = F + F^{\dagger}, \qquad A_1 = 0. \tag{56}$$

Because B is isotropic, $F^{\dagger} b = 0$ for $b \in B$, and

$$A_2 k = S, \qquad q_2(c, A_2 d) = 0 \quad (c, d \in C). \tag{57}$$

Thus the corrected tensor covariances

$$\widehat{\lambda}_2^{\pm} := \widehat{\lambda}_2'^{\pm} + A_2 \qquad \text{satisfy} \qquad k\widehat{\lambda}_1^{\pm} - \widehat{\lambda}_2^{\pm} k = 0. \tag{58}$$

The common smooth correction is invisible to the physical state and repairs the Ward identity exactly. This direct formula avoids an unproved global Witt normal form. It is a project-level composition of the Fredholm data in [16] and the abstract subsidiary/BRST mechanism in [18, 28].

6.3 Pairing-preserving transport to the five-field matrix

In grouped orders

$$\mathrm{WZ} : (u, b_W, c, \bar{c}), \qquad \text{source} : (\Theta, \beta, \eta, \bar{\eta}), \tag{59}$$

define configuration and Euler-row maps

$$C_0 = \mathrm{diag}(T_g, -1, 1, -1), \qquad R_0 = \mathrm{diag}(-\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}). \tag{60}$$

With the declared graded forms B_{WZ} and B_Q, one has

$$P_Q = R_0 L_{\mathrm{WZ}} C_0, \qquad B_Q R_0 = \frac{1}{2} C_0^* B_{\mathrm{WZ}}. \tag{61}$$

The second identity is the missing pairing statement: R_0 is not simply C_0^*. If $\lambda^{\pm}_{\mathrm{WZ}}$ is the exact host tensor/vector BRST pair, set

$$\lambda^{\pm}_Q = C_0^{-1} \lambda^{\pm}_{\mathrm{WZ}} R_0^{-1}. \tag{62}$$

Then

$$P_Q \lambda^{\pm}_Q = \lambda^{\pm}_Q P_Q = 0, \qquad \lambda^{+}_Q - (-1)^{\mathrm{gh}} \lambda^{-}_Q = i G_Q, \tag{63}$$

and the associated forms are Hermitian and positive on the transported physical kernel. The symbol $(-1)^{\mathrm{gh}}$ denotes the entrywise Schur sign.

The configuration codifferential and its pairing adjoint are

$$\gamma_Q(\Theta, \beta, \eta, \bar{\eta}) = (\bar{K}\eta, 0, 0, \beta), \qquad \gamma_Q^{\star}(\Theta, \beta, \eta, \bar{\eta}) = (0, \bar{\eta}, -D\Theta, 0). \tag{64}$$

Their exact Ward compatibility follows from the repaired tensor/vector pair. Add the scalar covariance as a BRST-inert spectator, restrict to U_F, permute to $(\Theta, \eta, \bar{\eta}, \beta)$, and transport through J_{nm}.

At unit $\hbar_{\mathrm{b}}$, the local benchmark operator pair is

$$\lambda^{\pm}_{\mathrm{b},1} = c_0 J_{\mathrm{nm}}^{-1} \lambda^{\pm}_H J^{\vee}_{\mathrm{nm}}, \qquad c_0 = \frac{\kappa^2}{4}, \qquad J^{\vee}_{\mathrm{nm}} = (J^{\dagger}_{\mathrm{nm}})^{-1}. \tag{65}$$

Equivalently, the dimensionful form is

$$\Lambda^{\pm}_{\mathrm{b}}(f, g) = \hbar_H \Lambda^{\pm}_{H,1}(J^{\vee}_{\mathrm{nm}} f, J^{\vee}_{\mathrm{nm}} g), \qquad \hbar_H = c_0 \hbar_{\mathrm{b}}. \tag{66}$$

One uses either equation (65) followed by $\hbar_{\mathrm{b}}$ or equation (66); applying both c_0 factors would double count the action normalization.

Theorem 6.2 (Full free-local BRST Hadamard system)**.** *At $\lambda = 1/2$, the host construction and equations (60) to (62) and (65) define a Hadamard pair for the complete field multiplet $(h, \phi, c, \bar{c}, b)$ on U_F. It has the exact five-field Green-matrix causal difference, obeys both field equations and the free BRST Ward identity, and is positive semidefinite on the ghost-number-zero physical kernel.*

This theorem contains no antifield two-point function. It also does not yet say that the physical form has no extra null class. That cohomological question is settled in section 8.

7 Equicausal free field algebra and time slice

7.1 Why the unrestricted microcausal class is not used

For $n \geq 1$, let

$$\Xi_n = (T^* U_F^n \setminus 0) \setminus (\overline{V}_+^n \cup \overline{V}_-^n). \tag{67}$$

The usual microcausal condition asks $\mathrm{WF}(F^{(n)}(\Phi)) \subset \Xi_n$ at each configuration. This pointwise condition is insufficient: a Peierls bracket of regular functionals can fail to be smooth, so the unrestricted class is not closed [22].

We instead use the graded equicausal space $\mathcal{F}_{\mathrm{ec}}^{\mathrm{gr}}$. Its elements have finite algebraic ghost and antighost degrees, are Bastiani smooth in (h, ϕ, b), have compactly supported mixed derivatives, obey the same wavefront exclusion, and satisfy the equicontinuity condition of [22] uniformly on compact subsets of the even configuration space. At fixed contraction order, the five internal species produce only a finite sum of ordinary bundle-component contractions. Koszul permutations alter signs but not wavefront sets. The source estimates therefore extend componentwise to the finite graded multiplet. This is a source-specialized analytic step, not a claim about a completed exterior topology.

7.2 Contraction kernel and star product

Let W^+ and W^- be the form-level members of the full Hadamard pair, including the Schur sign in the minus member. To avoid silently assuming a crossing identity not contained in the state theorem, define

$$W = \frac{1}{2}\left(W^+ + \tau_{\mathrm{gr}} W^-\right), \tag{68}$$

where τ_{gr} exchanges arguments and inserts the graded species sign. Ordinary Hermitian conjugation is complex conjugation followed by ungraded argument exchange; reversal of the exterior coefficient product already supplies the odd signs. Then

$$W^\dagger = W, \qquad W - \tau_{\mathrm{gr}} W = i E_G. \tag{69}$$

The symmetrized kernel is a Hadamard contraction with the correct commutator. It is not asserted to be the selected positive Hadamard covariance.

For homogeneous equicausal functionals define

$$F \star_W G = m_{\mathrm{gr}} \exp(\hbar_{\mathrm{b}} \Gamma_W)(F \otimes G) = \sum_{n \geq 0} \frac{\hbar_{\mathrm{b}}^n}{n!} \left\langle F_l^{(n)}, W^{\otimes n} G_r^{(n)} \right\rangle_{\mathrm{gr}}. \tag{70}$$

Every coefficient is equicausal, and the product is associative [22]. Equations equations (69) and (70) give

$$(F \star_W G)^\dagger = G^\dagger \star_W F^\dagger, \qquad [F, G]_{\star,\mathrm{gr}} = i\hbar_{\mathrm{b}} \{F, G\}_0 + O(\hbar_{\mathrm{b}}^2). \tag{71}$$

For linear functionals the commutator formula is exact.

7.3 Exact Ward identity and BRST descent

Let Π act as $+1$ on even and -1 on odd species. In grouped source order, the Hadamard covariances satisfy

$$\gamma_Q \lambda_Q^+ = \lambda_Q^+ \gamma_Q^\star =: M, \qquad \Pi M = -M \Pi. \tag{72}$$

Therefore

$$\Pi \gamma_Q \lambda_Q^+ + \lambda_Q^+ \gamma_Q^\star \Pi = 0. \tag{73}$$

The signed minus member and the symmetrization equation (68) obey the same two-argument identity. This is precisely the covariance condition needed for the unconjugated free BRST differential to be a star derivation [27].

Let s_0 act as in equation (14). It preserves the finite graded equicausal domain and

$$s_0(F \star_W G) = (s_0 F) \star_W G + (-1)^{|F|} F \star_W (s_0 G). \tag{74}$$

Let δ_P be the Koszul differential given by contraction with the linear equation section $P^{\mathrm{b}}_{\mathrm{nm},1/2}\Phi$. The exact covariance relation implies

$$[s_0, \delta_P]_{\mathrm{gr}} = 0. \tag{75}$$

The on-shell algebra is the degree-zero Koszul homology

$$\mathcal{A}_{\mathrm{ec,on}}(U_F) = H_0(\mathcal{X}^{\mathrm{gr},*}_{\mathrm{ec}}[[\hbar_{\mathrm{b}}]], \delta_P), \tag{76}$$

not merely a quotient by the algebraic ideal generated by constant linear equations. The full Koszul image is a two-sided star ideal. The free physical product algebra is

$$\mathcal{A}_{\mathrm{phys,free}} = H^0(\mathcal{A}_{\mathrm{ec,on}}(U_F), s_0). \tag{77}$$

7.4 Einstein causality and fixed-ambient time slice

If homogeneous F, G have causally disjoint functional supports, every cross contraction with the causal kernel vanishes. Equation equation (69) then implies, term by term,

$$F \star_W G = (-1)^{|F||G|} G \star_W F. \tag{78}$$

Let $N \subset U_F$ be open and contain a Cauchy surface of U_F. The Green-hyperbolic time-slice projection and its chain homotopy preserve equicausality [22]. Inclusion of the fixed-ambient support subcomplex inside N is therefore a quasi-isomorphism and induces

$$\mathcal{A}_{\mathrm{ec,on}}(N) \simeq \mathcal{A}_{\mathrm{ec,on}}(U_F), \qquad H^0(\mathcal{A}_{\mathrm{ec,on}}(N), s_0) \simeq H^0(\mathcal{A}_{\mathrm{ec,on}}(U_F), s_0) \tag{79}$$

as star and product algebras, respectively. The first is compatible with the ordinary dagger before BRST cohomology. The second is deliberately stated only as a product isomorphism; the dagger obstruction below prevents a stronger claim in the frozen convention.

Renormalized causal factorization is more restricted. Epstein–Glaser maps exist for compactly supported local, fiberwise analytic inputs and their finite multilocal products [8, 9, 19]. For causally ordered supports they obey

$$T_n(F_1, \dots, F_n) = T_k(F_1, \dots, F_k) \star_W T_{n-k}(F_{k+1}, \dots, F_n). \tag{80}$$

This source-specialized local/multilocal row does not extend T_n to every equicausal functional. Nor does it by itself descend through the equation ideal or BRST cohomology; collision terms and local anomalies remain.

8 Physical reduction, local injectivity, and the dagger obstruction

8.1 Nonminimal contraction

Before gauge fixing, the nonminimal jets obey

$$s_0\bar{c} = b, \quad s_0 b = 0, \quad s_0 b^{\ddagger} = -\bar{c}^{\ddagger}, \quad s_0 \bar{c}^{\ddagger} = 0. \tag{81}$$

Define the odd right derivation r by

$$r(D_I b) = D_I \bar{c}, \qquad r(D_I \bar{c}^{\ddagger}) = -D_I b^{\ddagger}, \qquad r(D_I \bar{c}) = r(D_I b^{\ddagger}) = 0. \tag{82}$$

If N_{nm} counts nonminimal jets, then

$$s_0 r + r s_0 = N_{\mathrm{nm}}. \tag{83}$$

Division by positive homogeneous degree gives a support-preserving contracting homotopy. For finite-jet fiberwise analytic germs the equivalent scaling integral is

$$H_{\mathrm{nm}} = r \int_0^1 \frac{\mathrm{d}t}{t} (\sigma_t - \iota\pi), \qquad s_0 H_{\mathrm{nm}} + H_{\mathrm{nm}} s_0 = 1 - \iota\pi. \tag{84}$$

Conjugating equation (84) by the canonical gauge-fixing transformation gives the corresponding gauge-fixed contraction. The transported minimal projection is not naive deletion of $\bar{c}, b$ and their antifields.

At ghost number zero, compact linear BV cohomology is consequently the reduced test-space phase equation (35). This exact local contraction is a project specialization of the standard nonminimal-doublet mechanism [5].

8.2 No extra null vectors on the host

The physical gravitational Cauchy space is

$$V_g = C/B, \qquad q_g([c],[d]) = q_2(c,d). \tag{85}$$

Equation equation (52) makes q_g weakly nondegenerate. Add the ordinary nondegenerate scalar phase space and write $V_{M_\Gamma} = V_g \oplus V_\phi$.

Let $\Lambda^{\pm}_{M_\Gamma}$ be the induced positive physical two-point forms. Their sum

$$N_{M_\Gamma} = \Lambda^{+}_{M_\Gamma} + \Lambda^{-}_{M_\Gamma} \tag{86}$$

is positive definite. If $N_{M_\Gamma}(v,v) = 0$, positivity and Cauchy–Schwarz make both $\Lambda^{\pm}_{M_\Gamma}(w,v)$ vanish for all w. Their causal difference then gives $q_{M_\Gamma}(w,v) = 0$ for all w, and nondegeneracy implies $v = 0$. Wrochna–Zahn's criterion therefore yields a ghost-number-zero pre-Hilbert space with no BRST-closed, nonexact zero-norm vector on the host [28].

8.3 Injectivity of local zero extension

The local state is a pullback along the physical map

$$I_{\mathrm{phys}} : V_{U_F} \longrightarrow V_{M_\Gamma}, \qquad [f] \longmapsto [i_* f]. \tag{87}$$

Its covariance radical is exactly $\operatorname{Ker} I_{\mathrm{phys}}$. Thus host positivity does not by itself prove local faithfulness.

Choose U_F to be a standard contractible double cone, diffeomorphic to $\mathbb{R}^4$. For the Calabi complex on a constant-curvature spacetime, causally restricted cohomology gives

$$H^1_{\mathrm{Cal,sc}}(U_F) \simeq H^2_{\mathrm{Cal},c}(U_F) = 0 \tag{88}$$

under the stated topology [25]. Hence every spacelike-compact pure-gauge metric perturbation admits a spacelike-compact gauge parameter on U_F.

Suppose $[f] \in V_{U_F}$ maps to zero. The host test/solution quotient theorem [21] gives a spacelike-compact ϵ_{M_Γ} with

$$G_{M_\Gamma} i_* f = K_{M_\Gamma} \epsilon_{M_\Gamma}. \tag{89}$$

Restriction of Green operators to a causally compatible region [2] gives

$$G_{U_F} f = K_{U_F}(i^* \epsilon_{M_\Gamma}). \tag{90}$$

The left side is spacelike compact even if $i^*\epsilon_{M_\Gamma}$ need not be. By equation (88) it equals $K_{U_F}\epsilon_{U_F}$ for a spacelike-compact local parameter. Injectivity of the local test/solution quotient then implies $[f] = 0$. Therefore

$$\operatorname{Ker} I_{\text{phys}} = 0. \tag{91}$$

Theorem 8.1 (Local no-extra-null physical space)**.** *On the selected standard double cone U_F, the ghost-number-zero free BRST physical form is positive definite at fixed $\hbar_{\text{b}} > 0$. A BRST-closed one-particle vector has zero norm if and only if it is BRST exact.*

This conclusion is narrower than Peierls nondegeneracy. A distinct adjoint Calabi parametrizability problem retains local Killing data, so the state-independent presymplectic radical and an independently functorial net are not promoted here.

8.4 The ordinary dagger does not descend

The ordinary dagger is valid on the off-shell and on-shell star algebra. It fails on the same s_0 cohomology. Define the adjoint differential on a homogeneous functional by

$$s_\sharp F = (-1)^{|F|}\big(s_0(F^\dagger)\big)^\dagger. \tag{92}$$

The minimal and nonminimal pieces satisfy

$$s_\sharp = s_{\min} - s_{\text{nm}}, \tag{93}$$

so dagger is an anti-isomorphism between the s_0 and $s_\sharp$ complexes, not an involution of the s_0 complex.

For an explicit counterexample choose real linear smearings for which the on-shell classes $C = s_0 H$ and $B = s_0 \ell_{\bar{c}}$ are nonzero and satisfy $C \star B \neq 0$. Such choices exist in the nontrivial free ghost–auxiliary polynomial sector; this nonvanishing condition is essential and is not asserted for arbitrary smearings. Abbreviate

$$H = \ell_h, \qquad C = s_0 H, \qquad B = s_0 \ell_{\bar{c}}. \tag{94}$$

The ghost-number-minus-one element $A = \ell_{\bar{c}} \star H$ has boundary

$$F = s_0 A = B \star H - \ell_{\bar{c}} \star C. \tag{95}$$

Ordinary reversal gives $F^\dagger = H \star B - C \star \ell_{\bar{c}}$ and hence

$$s_0(F^\dagger) = 2C \star B \neq 0. \tag{96}$$

Thus dagger sends an explicit boundary outside the cycle space.

Theorem 8.2 (Dagger-descent obstruction; derived)**.** *In the frozen nonminimal reality convention, the ordinary star-algebra dagger does not descend to* $H^0(\mathcal{A}_{\mathrm{ec,on}}, s_0)$*.*

One may seek an adjoint-BRST double-kernel construction or transfer a dagger from a reduced physical algebra after nonminimal contraction. Neither construction is provided here. Accordingly the physical object in equation (77) is presently a product algebra, while the no-extra-null theorem is a one-particle/pre-Hilbert statement.

9 Matter deformation and the quantum stop line

9.1 The exact $h\phi\phi$ vertex

Expanding the scalar action using

$$g^{\mu\nu} = \bar{g}^{\mu\nu} - \kappa h^{\mu\nu} + O(\kappa^2), \qquad \mathrm{d}\mu_g = (1 + \tfrac{\kappa}{2}h)\mathrm{d}\mu_{\bar{g}} + O(\kappa^2), \tag{97}$$

gives

$$S_m[\bar{g} + \kappa h, \phi] = S_{\phi,0} + \kappa V_{h\phi\phi} + O(\kappa^2), \tag{98}$$

where

$$\begin{aligned} V_{h\phi\phi} &= \int_{U_F} \left[\frac{1}{2}h^{\mu\nu}\bar{\nabla}_\mu\phi\bar{\nabla}_\nu\phi - \frac{1}{4}h((\bar{\nabla}\phi)^2 + m^2\phi^2)\right]\mathrm{d}\mu_{\bar{g}} \\ &= \frac{1}{2}\int_{U_F} h_{\mu\nu}T_0^{\mu\nu}[\phi]\mathrm{d}\mu_{\bar{g}}. \end{aligned} \tag{99}$$

Direct differentiation yields the off-shell identity

$$\bar{\nabla}^\mu T^0_{\mu\nu} = (P_\phi\phi)\bar{\nabla}_\nu\phi. \tag{100}$$

This is exactly the compatibility needed by the linearized Bianchi identity on the free scalar shell.

9.2 Scalar-antifield completion

The first nonlinear scalar transformation is $s_1\phi = \mathcal{L}_c\phi$. Define

$$A_\phi = \int_{U_F} \phi^\ddagger c^\mu\bar{\nabla}_\mu\phi, \qquad S_{1,m} = V_{h\phi\phi} + A_\phi. \tag{101}$$

The antifield is density valued, so no second volume density appears. With the frozen left/right derivative convention,

$$\frac{\delta_l A_\phi}{\delta\phi^\ddagger} = +\mathcal{L}_c\phi, \qquad \frac{\delta_r A_\phi}{\delta\phi^\ddagger} = -\mathcal{L}_c\phi. \tag{102}$$

Using equation (100) and compact support,

$$(V_{h\phi\phi}, S_{\mathrm{BV},0}) = -\int (P_\phi\phi)c^\nu\bar{\nabla}_\nu\phi\,\mathrm{d}\mu_{\bar{g}}, \tag{103}$$

whereas

$$(A_\phi, S_{\mathrm{BV},0}) = +\int (P_\phi\phi)c^\nu\bar{\nabla}_\nu\phi\,\mathrm{d}\mu_{\bar{g}}. \tag{104}$$

Therefore

$$(S_{\mathrm{BV},0}, S_{1,m}) = 0. \tag{105}$$

Theorem 9.1 (First matter BV subcocycle; derived)**.** *The $h\phi\phi$ vertex with coefficient and sign in equation (99), together with the scalar-antifield term in equation (101), is a closed first-order deformation of the free BV complex.*

At $\bar{\phi} = 0$ the free Hessian is block diagonal and the free BV differential preserves scalar field-content degree, so this is a separately closed matter summand. It does not audit the hhh, metric-antifield, ghost-antifield, or gauge-fixing/ghost summands of the full order-κ action. It also says nothing about $(S_{1,m}, S_{1,m})$ at order κ^2.

With a compact switching function f,

$$(S_{1,m}(f), S_{\mathrm{BV},0}) = -\int (\bar{\nabla}_\mu f) T_0^{\mu\nu} c_\nu \, \mathrm{d}\mu_{\bar{g}}, \tag{106}$$

which vanishes in an observation region contained in the plateau $f = 1$.

9.3 One renormalized stress insertion

Hollands and Wald show that, in dimension greater than two, the time-ordered-product axioms can be supplemented so that the renormalized scalar stress tensor is conserved [23]. Choose such a prescription and define the single factorized insertion

$$V_R(f) = \frac{1}{2}\int f h_{\mu\nu} T_R^{\mu\nu} \mathrm{d}\mu_{\bar{g}}. \tag{107}$$

The free graviton–scalar covariance is block diagonal, so this definition introduces no h–ϕ contraction. The field-only free BRST differential obeys

$$s_0^{\mathrm{fld}} V_R(f) = -\int (\bar{\nabla}_\mu f) c_\nu T_R^{\mu\nu} \mathrm{d}\mu_{\bar{g}}. \tag{108}$$

This is a source-specialized one-insertion, cutoff-plateau Ward statement. It is not the off-shell BV anomalous Master Ward identity.

9.4 Typed AMWI/QME obstruction

Enlarge the local input space only for this obstruction problem by treating antifields as external sources, with no antifield propagator. The renormalized BV anomaly map has the type [10, 15]

$$\Delta_r : \mathcal{D}^{\mathrm{even},0}_{T,\mathrm{BV}} \longrightarrow \mathcal{A}^{\mathrm{odd},1}_{\mathrm{loc},\mathrm{BV}}[[\hbar_{\mathrm{b}}]], \qquad \operatorname{supp}\Delta_r(V^T) \subset \operatorname{supp} V^T. \tag{109}$$

Select a free BRST-compatible normalization at the quadratic base point. For $V_m(f) = \kappa S_{1,m}(f)$, the first possible matter quantum obstruction is the joint $\kappa\hbar_{\mathrm{b}}$ coefficient. Pulling it back through the Wick map defines a local four-form representative $a_{1,1}^{(m,4)}$ obeying the consistency condition

$$s_0^{\mathrm{BV}} a_{1,1}^{(m,4)} + \mathrm{d}_H b_{1,1}^{(m,3)} = 0. \tag{110}$$

It therefore determines a restricted local cohomology class

$$[a_{1,1}^{(m,4)}] \in H^{1,4}_{\mathrm{loc},\mathcal{R}}(s_0^{\mathrm{BV}} \mid \mathrm{d}_H). \tag{111}$$

Allowed finite renormalizations shift the representative by an s_0^{BV}-exact plus horizontally exact term. Thus vanishing of equation (111) is the precise first matter anomaly-removal gate within the declared renormalization class.

The nonminimal quartet can be removed from this raw restricted cohomology by the support-preserving analytic homotopy equation (84), transported through the gauge fermion. Several minimal orientation-even, antifield-zero candidate slices are also ambient-exact. These reductions do not prove that the actual anomaly-map image lands in those slices.

Excluded filtration claim. An attempted finite source-filtration argument proposed an additional bound after the nonminimal projection. In the gauge-fixed variables, however, the canonical minimal projection is not naive substitution, and the claimed postprojection degree bound has not survived audit. That argument and its checker are not used anywhere in this manuscript. No conclusion about equation (111) is drawn from it.

The full nonlinear local BRST classifications [4, 5] provide strong comparison evidence that four-dimensional Einstein gravity with ordinary matter has no local diffeomorphism anomaly of the familiar chiral type. They do not compute equation (111): the latter uses a background-linearized differential, AdS reducibility data, a restricted scaling/local-covariance class, and a selected anomaly map. Global and boundary anomalies are also outside those local comparisons.

10 What the benchmark does and does not establish

10.1 Compatibility achievements

The benchmark establishes a nontrivial same-domain compatibility chain:

(i) a classical nonminimal BV datum and exact classical master equation;
(ii) a complete five-field Green matrix, including $\lambda = 0$;
(iii) a positive reduced graviton–scalar Hadamard state on a selected local domain;
(iv) an exact tensor/vector Ward completion and normalization into the nonminimal BRST matrix;
(v) a five-field equicausal formal star algebra with the correct commutator;
(vi) a positive no-extra-null ghost-number-zero one-particle space;
(vii) free Einstein causality and fixed-ambient time slice;
(viii) an exact first matter deformation subcocycle and a typed first anomaly class.

These items are useful as acceptance tests for boundary reconstruction programs: a candidate boundary sector that cannot reproduce this local chain has not yet matched a standard perturbative gravity benchmark.

10.2 The assumed-dynamics limitation

The Einstein–scalar action equation (4) is an input. The benchmark does not derive it from a positive boundary functional, a sewing axiom, a moment problem, or a microscopic amplitude. It does not select G, Λ, m, the matter content, or the background. Hence the work is properly called a compatibility benchmark, not a new dynamical law of quantum gravity.

10.3 Why local AdS is not global AdS

The Fuchsian host is boundaryless and globally hyperbolic. It is not global AdS with reflecting conditions. A global promotion would require one common domain for $(h, \phi, c, \bar{c}, b)$ satisfying at least:

- **B1:** BRST-invariant boundary conditions;
- **B2:** separation of proper gauge ghosts from asymptotic symmetries and charges;
- **B3:** advanced/retarded boundary Green operators with the Ward identity;
- **B4:** vanishing total graviton, scalar, and ghost symplectic flux;
- **B5:** a positive physical boundary Hadamard state;
- **B6:** a boundary-compatible causality and time-slice replacement;
- **B7:** GHY, holographic, gauge-fixing, ghost, and BV–BFV boundary terms in one master equation;
- **B8:** bulk-plus-boundary renormalization and anomaly control;
- **B9:** a nonlinear relational or boundary dressing and operational readout.

Classical mode analyses [24], scalar AdS states [12], and classical BV–BFV gravity [11] each address important parts. None supplies the complete coupled Lorentzian graviton–scalar–ghost package used by the list.

10.4 Exact stop line

The following implications are not licensed by the present results:

No full order-κ classical deformation has been audited. No interacting quantum master equation, interacting BRST cohomology, or anomaly cancellation theorem has been proved. No global reflecting AdS boundary theory, operational boundary instrument, continuum regulator path, emergent metric derivation, infrared Einstein recovery theorem, ultraviolet completion, or empirical prediction follows.

(112)

Even within the free theory, the ordinary dagger is absent on the frozen BRST cohomology and the independently functorial net remains open. These are not minor typographical gaps: they delimit the mathematical object that has actually been constructed.

11 Discussion

11.1 What is genuinely learned

The central lesson is that a plausible verbal stack—"BV plus Hadamard plus BRST plus pAQFT"—is not a theorem. Compatibility requires a series of maps. The source-to-benchmark action normalization fixes the sign of the Euler operator. The Gérard-to-source bridge contributes a single factor one half. The Wrochna–Zahn configuration and Euler-row maps are dual only after the pairing identity equation (61). The nonminimal field matrix is Green hyperbolic even when the Schur complement is undefined. Host positivity reaches the local algebra only after zero-extension injectivity. Finally, the same ghost convention that makes the covariance and BRST Ward identity work can obstruct descent of the ordinary dagger.

These are modest compared with constructing quantum gravity, but they are not empty bookkeeping. They identify where familiar theorems genuinely compose and where an additional physical or analytic input is needed.

11.2 Why the dagger failure is useful

The explicit counterexample equation (96) prevents a common silent move: declaring that because the off-shell star algebra and BRST complex each have reasonable reality properties, the cohomology automatically inherits the same adjoint. It does not. A transferred adjoint may still exist, but it must be constructed together with the nonminimal reduction or a physical completion. For boundary reconstruction this matters because positivity and operational interpretation require a specified involution, not merely a complex product algebra.

11.3 Interactions are the decisive next test

The classical matter subcocycle is a useful first deformation because its cancellation directly exposes the stress identity and the antifield sign. Yet it is only one summand. The next scientifically meaningful step is not another general criterion. It is to assemble the full order-κ BV deformation in one convention, construct renormalized time-ordered maps on its declared domain, and compute the first local anomaly class. Only after that calculation can the free benchmark be promoted to an interacting local sector.

12 Conclusion

We have assembled a self-contained local Einstein–scalar BV/BRST benchmark on a globally hyperbolic interior region of AdS_4. The construction reaches farther than a formal classical complex: it includes the full nonminimal Green matrix, a selected Hadamard host, exact Ward and normalization transport, an equicausal star algebra, local BRST no-extra-null positivity, causality, time slice, and a first matter deformation. It also records one exact obstruction—the ordinary dagger does not descend in the frozen nonminimal convention—and one exact stop condition: the first quantum matter anomaly class is typed but has not been computed.

The correct interpretation is therefore narrow but useful. This is a perturbative effective-field-theory benchmark for reconstruction architectures. It shows that a substantial compatibility stack can be realized locally once Einstein dynamics is assumed. It does not derive that dynamics, construct global AdS quantum gravity, or solve the interacting QME. The stop line equation (112) is part of the result.

A Convention dictionary

A.1 Curvature and signature

We use signature $(-,+,+,+)$ and

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma. \tag{113}$$

On AdS_4,

$$R_{\mu\nu\rho\sigma} = -L^{-2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}), \qquad R_{\mu\nu} = -3L^{-2}g_{\mu\nu}. \tag{114}$$

The source vector operator is $R = -\frac{1}{2}(\Box + \mathrm{Ric})$, while the local benchmark ghost equation is $M = \Box + \mathrm{Ric} = -2R$. These are convention-related blocks, not different dynamics.

A.2 Field, source, and Planck normalizations

The canonical perturbation is $g = \bar{g} + \kappa h$. Source variables are related by

$$\theta = \frac{\kappa}{4}(h - \tfrac{1}{2}\bar{g}\ \mathrm{tr}\, h), \qquad \zeta = \frac{\kappa}{2}\phi. \tag{115}$$

The action Hessian and causal form obey

$$P_{0,\mathrm{b}} = -\frac{4}{\kappa^2}\mathcal{A}^\dagger \bar{P}\mathcal{A}, \qquad \sigma_{\mathrm{b}} = \frac{\kappa^2}{4}\sigma_H \circ (J^\vee \times J^\vee). \tag{116}$$

Identified commutators require $\hbar_H = (\kappa^2/4)\hbar_{\mathrm{b}}$. The factor is applied once.

A.3 BRST configuration versus functional differential

The transported configuration codifferential has

$$\gamma_{\mathrm{b}}(H, c, \bar{c}, b) = (Kc, 0, -b, 0). \tag{117}$$

For a homogeneous evaluation functional ℓ we define the left functional action by the super-transpose rule

$$(s_0\ell)(x) = (-1)^{|\ell|}\ell(\gamma_{\mathrm{b}}x). \tag{118}$$

Thus the configuration component $\gamma\bar{c} = -b$ produces the functional rule $s_0\bar{c} = +b$. Flipping one field by hand would also conjugate the Euler matrix and covariances and is not an allowed shortcut.

B Direct block verification of the Green matrix

For convenience suppress the advanced/retarded sign and set

$$X = -E + \alpha\bar{K}C^2D, \quad Y = \bar{K}C, \quad Z = CD, \quad \alpha = \frac{1-2\lambda}{2}. \tag{119}$$

Then

$$B_\lambda \begin{pmatrix} X & Y \\ Z & 0 \end{pmatrix} = \begin{pmatrix} -\bar{P}X + \bar{K}Z & -\bar{P}Y \\ DX + \lambda Z & DY \end{pmatrix}. \tag{120}$$

Using $\bar{P}E = 1 - \bar{K}CD$, $\bar{P}\bar{K} = 0$, $D\bar{K} = R$, $RC = 1$ on compact sources, and $DE = \frac{1}{2}CD$, the diagonal blocks are identities. The lower-left block is

$$-\frac{1}{2}CD + \alpha CD + \lambda CD = \left(-\frac{1}{2} + \frac{1-2\lambda}{2} + \lambda\right) CD = 0. \tag{121}$$

The right inverse is analogous, using $C^2R = C$. For the odd block,

$$\begin{pmatrix} 0 & -R \\ R & 0 \end{pmatrix}\begin{pmatrix} 0 & C \\ -C & 0 \end{pmatrix} = I, \tag{122}$$

with the same result in the reversed order. This algebraic verification explains why the full matrix exists at $\lambda = 0$ although a Schur formula does not.

C Relative cohomology interpretation of local injectivity

Extension by zero is an injective chain map between compact linear BRST complexes,

$$0 \longrightarrow \mathcal{L}_c(U_F) \xrightarrow{i_*} \mathcal{L}_c(M_\Gamma) \longrightarrow \mathcal{L}_c(M_\Gamma, U_F) \longrightarrow 0. \tag{123}$$

The long exact sequence contains

$$H^{-1}(\mathcal{L}_c(M_\Gamma, U_F)) \xrightarrow{\delta_{\mathrm{rel}}} H^0(\mathcal{L}_c(U_F)) \xrightarrow{H^0(i_*)} H^0(\mathcal{L}_c(M_\Gamma)). \tag{124}$$

Under the reduced-phase identifications, $H^0(i_*) = I_{\mathrm{phys}}$, so

$$\operatorname{Ker} I_{\mathrm{phys}} = \operatorname{im} \delta_{\mathrm{rel}}. \tag{125}$$

The proof of equation (91) therefore shows that the connecting image vanishes; it does not require the relative group itself to vanish. The nonminimal relative summand is contractible, and the scalar summand is injective by the ordinary exact Green sequence. The nontrivial content is the gravitational support-localization argument using equation (88).

D Claim and provenance matrix

Claim	Status	Basis
Classical master equation	source-specialized	Standard BV theorem plus compact-support Einstein–scalar specialization.
Full five-field Green matrix	derived	Normal-hyperbolic source blocks plus explicit two-sided block inversion.
Reduced CCR/Weyl algebra	source-specialized	Linear gauge theory and Einstein–Klein–Gordon quotient theorems.
Fuchsian host embedding	mixed	Fuchsian AdS geometry plus finite bad-set and residual-finiteness refinement.
Reduced positive Hadamard state	mixed	Gérard gravity state, scalar Hadamard state, exact pairing and normalization pullback.
Full nonminimal Hadamard pair	derived composition	Fredholm left inverse, physical-state extension, Ward repair, Wrochna–Zahn matrix transport.
Equicausal star algebra	mixed	Equicausal closure theorem plus finite graded multiplet and exact local benchmark Ward identity.
Local no-extra-null BRST space	mixed	Host coisotropy and positivity plus Calabi zero-extension injectivity.
Ordinary dagger on BRST H^0	excluded	Explicit boundary whose dagger is not a cycle.
First matter subcocycle	derived	Direct expansion, stress divergence, and antifield-sign cancellation.
One-insertion stress Ward row	source-specialized	Hollands–Wald conserved renormalized stress prescription.

Claim	Status	Basis
$\kappa\hbar$ matter anomaly class	open	Class is typed in restricted local cohomology but not computed.
Full interacting QME	open	Remaining order-κ vertices and all collision/anomaly conditions absent.
Global reflecting AdS	open	No common graviton–scalar–ghost boundary theorem.
Emergent gravity or UV completion	not claimed	Einstein dynamics is assumed; no regulator/IR or nonperturbative bridge.

E Portable algebra verification

The packet contains four standalone SymPy programs for the block Green matrix, pairing-preserving Ward transport, graded functional signs and first matter vertex. Running `pythonreproduce.py` verifies respectively 7, 15, 17 and 10 printed algebra rows. These reproduce the previously checked coefficient calculations without depending on the original project. No program is claimed to prove Green support, Hadamard wavefront conditions, a Fredholm estimate, local-support cohomology, renormalization or an interacting QME. The excluded naive source-filtration checker is not bundled or counted as evidence. The analytic statements are conditional/source-specialized exactly where their PDE, microlocal or cohomological inputs enter. A complete fresh independent verification of every nested analytic bridge has not been performed.

Workflow and unresolved attribution

The operator reports that earlier work was led mainly by Fable 5, while results from July 2026 onward were predominantly produced by GPT models. DeepSeek and multiple GPT models also participated in review, checking, preparation and organization. This chronological account is not a complete audited allocation of individual contributions, and a later revision does not reattribute an earlier result. Human author names and affiliations are intentionally blank. These results have not undergone human expert review. Automated checks and AI reviews are further AI review, not human peer review. Historical source dates, this 29 October 2026 preparation date, and public disclosure dates are distinct. No authorship, ownership, publication-priority or journal-acceptance decision is made. The operator states:

> I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results.

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